

Intelligent Document Extraction, Validation & API Platform

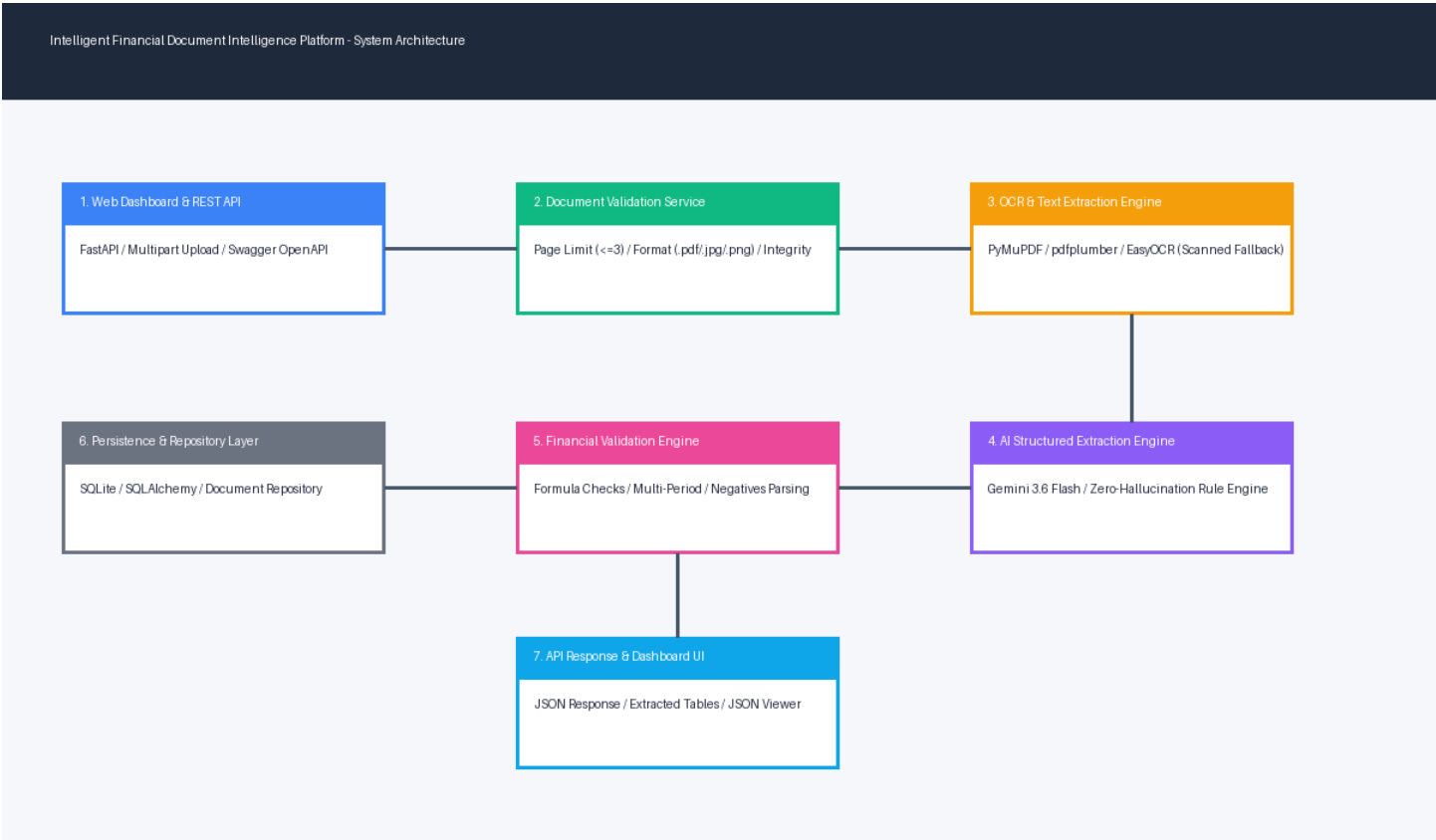
Technical Case Study Implementation Presentation

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1. Core Problem & Objective

Financial institutions process thousands of unstructured financial documents daily. This solution provides an end-to-end AI-powered document intelligence platform for 4 document categories: Invoices, Balance Sheets, Profit & Loss Statements, and Cash Flow Statements.

2. System Architecture



3. Pipeline & Key Capabilities

Component	Implementation Strategy & Technology
Document Validation	Boundary validation for page count (<=3), extensions (.pdf/.jpg/.png), 0-byte detection, & corruption check prior to OCR.
OCR & Text Extraction	Hybrid PyMuPDF / pdfplumber for native digital text & line layout; EasyOCR fallback for scanned PDFs/images.
AI Extraction Engine	Gemini 3.6 Flash LLM integration with zero-hallucination rule-based fallback. Missing values returned as null.
Financial Validation	Dedicated formula reconciliation engine evaluating independent comparative periods, bracketed negative numbers, & configurable tolerance.
Persistence Layer	SQLite database via SQLAlchemy ORM with repository pattern storing latest processing versions by document name.
REST API & UI Dashboard	FastAPI OpenAPI endpoints (POST /process, GET /{name}, GET /documents, GET /health) & Jinja2/Bootstrap HTML dashboard.

4. Verification & Automated Test Coverage

Comprehensive Pytest test suite covering 18 test cases across file boundary validation, multi-period financial formulas, OCR text extraction, zero-hallucination null handling, and full FastAPI integration tests with 100% pass rate.